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CBIM Problem Canvas

Project: Credit Pre-Screening Engine — Loan Intake Optimisation
Use Case: MUC02 — Path 3
Domain: Banking and Financial Services

CBIM Component	Problem Definition
1. Situation	The bank processes approximately **3,000 loan applications per month**. Each application requires around **25 minutes of full manual review**. Historical applicant information, credit behaviour, affordability indicators, and loan decisions are available for analysis.
2. Complication	Reviewing every application manually creates a significant workload and delays loan processing. Automating decisions could improve efficiency, but incorrectly declining eligible applicants or fast-tracking risky applicants creates financial, customer, and compliance risks. The bank therefore requires the observed wrongful-decline and wrongful-fast-track rates to remain within **5% for each automated lane**.
3. Question	How can the bank use machine learning to identify applications suitable for **auto-decline**, **fast-track approval**, or **full manual review**, while reducing analyst workload and keeping each automated lane's observed error rate within 5%?
4. SSOT (Single Source of Truth)	The historical loan application dataset containing **12,000 records**, including applicant income, credit score, existing obligations, loan amount, employment details, delinquency history, and historical approval/decline decisions. These recorded decisions serve as the modelling target and retrospective evaluation reference.
5. Success Definition	Develop and compare predictive models; classify applications into three review lanes; maintain observed wrongful-decline and wrongful-fast-track rates at or below **5% each**; and quantify automation, monthly reviews avoided, and potential analyst hours freed. Under the conservative Gradient Boosting experiment, the observed results were **36.67% automation**, **3.17% wrongful declines**, **4.00% wrongful fast-tracks**, and approximately **458.33 gross analyst hours freed per month**. These are retrospective results, not production guarantees.
6. Primary Stakeholder	**Bank Credit Operations Manager / Loan Processing Head**, responsible for reducing application-review workload while maintaining lending policy compliance, decision quality, and customer protection. Credit risk analysts and compliance teams are supporting stakeholders.

Proposed Business Approach

Use a **Gradient Boosting classification model** with two approval-probability thresholds to route applications into three lanes:

- **Auto-decline:** Applications with sufficiently low predicted approval probability.

- **Fast-track approval:** Applications with sufficiently high predicted approval probability, subject to verification.
- **Manual review:** Uncertain applications requiring assessment by credit analysts.

Final Recommendation: Pilot the conservative Gradient Boosting policy through fresh, preferably time-based shadow validation before enabling automated lending decisions. Monitor error rates, fairness across applicant groups, and actual processing-time savings.